

Eleana Karachaliou

ekarachal@gmail.com

Profile

I have successfully defended my PhD at the University of Manitoba (Biological Sciences) and will be graduating in October 2025. I am currently undertaking an internship (Environmental Junior Analyst) with the United Nations Environment Program (Latin America and the Caribbean Office) in the Science Policy Unit until October 31st, 2025. I have a background in Marine Biology and Data Science. During my PhD I worked on using open data to uncover human-caused and environmental drivers of global genetic diversity in marine fishes, and anadromous sea lamprey conservation genomics. I am interested in how climate change and human activity affect marine ecosystems, conservation planning and policies needed to prevent large-scale biodiversity declines, and promoting scientific outreach and education.

Education

2019 - October 2025	PhD, Department of Biological Sciences, Faculty of Science, University of Manitoba, Canada.
2023 (November)	GOOD-OARS-CLAP-COPAS Summer School. CEAZA & Universidad Católica del Norte, La Serena, Chile.
2017-2019	MSc Systems Biology, Department of Biotechnology, Agricultural University of Athens, Hellenic Republic (Grade: 9.19/10).
2018 (June)	Summer School in Aquatic Field Work, Technical University of Denmark (DTU).
2015	Erasmus exchange program (Spring Semester), BOKU, Austria.
2011-2017	Integrated BSc and MSc in Ichthyology and Aquatic Environment, Department of Agriculture, Ichthyology and Aquatic Environment, School of Agricultural Sciences, University of Thessaly, Hellenic Republic (Grade: 6.94/10).
2005-2011	Nikaia High School, Larisa, Hellenic Republic.

Employment History

May - October 2025	Intern (Environmental Junior Analyst) Science Policy Unit – United Nations Environment Program LAC Office, Panama City, Panama
2025	Instructor – Principles of Ecology (Laboratory), University of Manitoba (Winter Semester).
2021-2023	Teaching Assistant – Evolutionary Biology, University of Manitoba (Fall Semester).
2021-2023	Grader/Marker – Evolutionary Biology, University of Manitoba (Fall Semester).
2016-2017	English Teaching Assistant, Yellow Umbrellas Kindergarten, Athens, Hellenic Republic.
2015	Summer Internship, Multihull Yachting, Lavrio, Hellenic Republic. Assisting with the organization of trips, maintenance and transport of vessels, communication with ports and customers.
2014	Summer Internship, Piraeus Directorate of Fisheries, Athens, Hellenic Republic. Responsible for issuing aquaculture and fishing permits, archives, assisting with professional training and outreach programmes for fishermen.
2013-2015	Scientific Technician, Thales Project – Marine Angiosperm Tolerance on Environmental Stress (University of Thessaly). Collecting seagrass in the Pagasitikos and Sikia Gulf as part of a diving team and writing a report on species occurrence and distribution.

Publications

- **Karachaliou E.**, Schmidt C., de Greef E., Docker M.F., Garroway C.J. (2025) Urbanization is associated with reduced genetic diversity in marine fish populations. *Molecular Ecology*, 0:e17711
doi: <https://doi.org/10.1111/mec.17711>
- Schmidt C., **Karachaliou E.**, Vandergast A., Crandall E., Falgout J., Hunter M., Kershaw F., Leigh D., O'Brien D., Paz-Vinas I., Segelbacher G., Garroway C.J. (2025) A survey of mammal and fish genetic diversity across the global protected area network. *Conservation Letters*, 18:e13092
doi: <https://doi.org/10.1111/conl.13092>
- Paz-Vinas I., Vandergast A.G., Schmidt C., Leigh D.M., Blanchet S., Clark R.D., Crandall E.D., de Kort H., Falgout J., Garroway C.J., **Karachaliou E.**, Kershaw F., O'Brien D., Pinsky M.L., Segelbacher G., Toczydlowski R.H., Hunter M.E. (2025) Sparse genetic data limit biodiversity assessments in protected areas globally. *Frontiers in Ecology and Evolution*, e2867
doi: <https://doi.org/10.1002/fec.2867>

Research

- 2019 - 2025** Past & contemporary patterns of marine biodiversity: Making inferences from anadromous sea lamprey population genomics & publicly archived genetic data – PhD Thesis.
- 2018 - 2019** Biomonitoring of marine mammals in the Eastern Mediterranean – Participation in research cruises, English language narrator for short documentary produced as part of the project (University of Thessaly).

Aim: Population assessment for dolphins of the Corinthian Gulf – morphological identification of species and measurements for abundance and distribution.
- 2018 - 2019** DNA barcoding and phylogenetics of elasmobranch species in the Eastern Mediterranean – MSc thesis.
- 2018** Species identification using molecular techniques for sharks of the genus *Galeus* – laboratory rotation at the Molecular Biology Laboratory (Agricultural University of Athens) as part of MSc Degree.

Aim: DNA barcoding of the mitochondrial COI gene aiming to assign specimens to the species *Galeus melastomus*, confirming morphological data.
- 2015** Isolation & characterization of two *Hox* genes (A13a&A13b) in *Solea Solea* – Undergraduate thesis

Aim: Isolation and characterization of *Hox* genes via PCR amplification for the common sole *Solea solea*, a promising aquaculture species. Sequencing the two *Hox* genes led to identification of differences in gene expression and structure between viable and non-viable egg batches.

Workshops

- 2023 (October)** Workshop on conservation of the sea lamprey (*Petromyzon marinus*) – County Administrative Board in Halland, Swedish Agency for Marine and Water Management, County Administrative Boards in Skåne and Västra Götaland, SLU Swedish Species Information Center, Falkenberg, Sweden (Invited Speaker).
- 2022 (November)** CCFR – SCL workshop on Indigenous engagement (Online).
- 2020 (June/July)** Online workshop series on data management and synthesis in ecology and evolution – CIEE Living Data Project (Online).
- 2019 (November)** Genomic solutions for wildlife management and conservation in Canada – Genome Prairie (Online).
- 2019 (October)** Lamprey genomics workshop, University of Manitoba, Winnipeg, Canada.
- 2019 (February)** Workshop on molecular evolution: Patterns and causes – Konnevesi Research Station of the University of Jyväskylä, Finland.

Technical Skills

- Programming languages: R, bash command line (Linux/Unix)
- Working with High-Performance Computing
- Bayesian generalized linear mixed modelling
- Working with open data and data synthesis
- Familiar with Python, ArcGIS and QGIS
- Microsoft Office

Languages

- Bilingual English and Greek
- Spanish – Beginners level
- Anishnaabemowin/Ojibwe – Beginners level

Other Qualifications

- IELTS English Certificate
- PADI Open Water Diving Certificate

Awards

- Department of Biological Sciences & Faculty of Science Travel Award, University of Manitoba (2024, 2023, 2022)
- Faculty of Graduate Studies Travel Award (2024)
- Faculty of Graduate Studies Research Completion Scholarship Award (2023)
- ICES PICES Early Career Scientist Conference 2022 Best Poster Award (2022)
- MEOPAR Conference Travel Award (2022)
- International Graduate Student Entrance Scholarship, University of Manitoba (2019)

First Author Conference Presentations

- Karachaliou E., Schmidt C., de Greef E., Docker M.F., Garroway C.J. (2024) Urbanization drives genetic diversity declines in marine fish populations globally. IMCC7, (Oral) October 2024, Cape Town, South Africa.
- Karachaliou E., Suurväli J., de Greef E., Müller C., Garroway C.J., Docker M.F. (2024) Native sea lamprey evolutionary history and conservation implications. AFS, (Oral) September 2024, Honolulu, Hawaii.
- Karachaliou E., de Greef E., Suurväli J., Müller C., Garroway C.J., Docker M.F. (2024) Past, present & future sea lamprey (*Petromyzon marinus*) conservation genomics. SBE Annual Meeting, (Poster) July 2024, Puerto Vallarta, Mexico.
- Karachaliou E., Suurväli J., Evanno G., Castro-Santos T., Wilkie M., Jóhannsson M., Jónsson B., Baer J., Bracken F., Mateus C., Svensson M., Thysell E., Evans T., Fisk M., Hogue G.M., Garroway C.J., Docker M.F. (2023) Using whole genome re-sequencing to uncover population structure in anadromous sea lamprey across its native range. CSEE – CBA Joint Meeting, (Oral) June 2023, Winnipeg, Canada.
- Karachaliou E., Schmidt C., Docker M.F., Garroway C.J. (2022) Environmental & human disturbance effects on global marine genetic diversity. ESA & CSEE Joint Meeting, (Oral) August 2022, Montreal, Canada.
- Karachaliou E., Garroway C.J., Grayson P., Evanno G., Docker M.F. (2022) Whole genome re-sequencing uncovers historic transoceanic divergence in anadromous sea lamprey *Petromyzon marinus* – implications for conservation & management. Fourth ICES PICES Early Career Scientist Conference, (Poster) July 2022, St. John's, Canada.
- Karachaliou E., Schmidt C., Docker M.F., Garroway C.J. (2021) Global drivers of genetic diversity in marine fish species. CSEE Virtual Meeting, (Oral-Virtual) August 2021.
- Karachaliou E., Garroway C., Grayson P., Evanno G., Docker M.F. (2021) Reconstructing anadromous sea lamprey evolutionary history – implications for management & conservation. (Oral – Virtual) CCFR-SCL Virtual Meeting, February 2021.

Peer-reviewed Conference Presentations

- Karachaliou E., Hatzopoulos P., Exadactylos A., Gkafas G.A., Megalofonou P. (2018) Cytochrome Oxidase Subunit I (COI) Gene - A method for species identification of the blackmouth catshark (*Galeus melastomus*). HydroMediT, November 2018, Volos, Hellenic Republic.

Seminars

- Department of Biological Sciences Crackerjack Seminar Series, University of Manitoba (January 2024) Title: Effects of human activity & oceanographic conditions on global marine fish species genetic diversity. Winnipeg, Canada.
- CRITFC Brown Bag Seminar Series (August 2022) Title: Implications of marine genetic diversity patterns on conservation & management: What can we learn from anadromous sea lamprey & publicly archived genetic data? Portland, USA.

Webinars

- Sea Lamprey Webinar (Plenary presentation) – County Administrative Board of Halland County, Swedish Agency for Marine and Water Management and SLU Swedish Species Information Centre, Online (November 2024).
- Indigenous Engagement Session: An information, engagement session on lampricide treatments and concerns in the Laurentian Great Lakes. (Guest speaker) – Mississauga First Nation, Online (June 2022).
- Sea Lamprey Webinar (Invited speaker) – County Administrative Board of Halland County, Swedish Agency for Marine and Water Management and SLU Swedish Species Information Centre, Online (April 2021).

Other Experiences

- Presentations at annual Department of Biological Sciences Darwin Day events, University of Manitoba (2020-2024).
- Session Chair, CSEE – CBA Joint Meeting, Winnipeg, Canada (June 2023).
- Member of Search Committee for the position of Assistant Professor, Department of Biological Sciences, University of Manitoba (October - December 2022).
- Poster Session Judge, MEOPAR Annual Networking Meeting, Online (November 2022).
- Honor the Wykanish Event (August 2022): Participation in community outreach event with Yakama Nation Fisheries, Prescott, WA, USA.
- Research Visit – Columbia River Inter-Tribal Fish Commission & Yakama Nation Fisheries, Portland OR & Toppenish WA, USA (Summer 2022). Field work with Pacific lamprey, fin clips for DNA sampling, electrofishing, fish handling, tagging, habitat surveys, meeting with genetic monitoring expert.
- Session Moderator, CSEE-SCEE Online Conference (August 2021).
- Interpreter at the 1st Hydromedit International Congress 2014 for the session: Round Table on Management Difficulties of Lake Karla, Volos, Hellenic Republic.
- Participation in the 20th European Youth Parliament National Selection Conference 20-22 November 2009, Thessaloniki, Hellenic Republic.

Referees

- Dr. Margaret F. Docker, Professor, Department of Biological Sciences, University of Manitoba, Winnipeg, Canada (PhD Supervisor) email: Margaret.Docker@umanitoba.ca
- Dr. Colin J. Garroway, Associate Professor, Department of Biological Sciences, University of Manitoba, Winnipeg, Canada (PhD co- Supervisor) email: Colin.Garroway@umanitoba.ca
- Cole Robson-Hyska, Instructor I, Department of Biological Sciences, University of Manitoba, Winnipeg, Canada (Supervisor during Instructor – Principles of Ecology laboratory course teaching position) email: Cole.Robson-Hyska@umanitoba.ca